

A2: TEC Heating and Cooling Analysis

Ricky Huang and Xavier Zhu | Phys 39 | Module 4

Steady temperature vs signed PWM

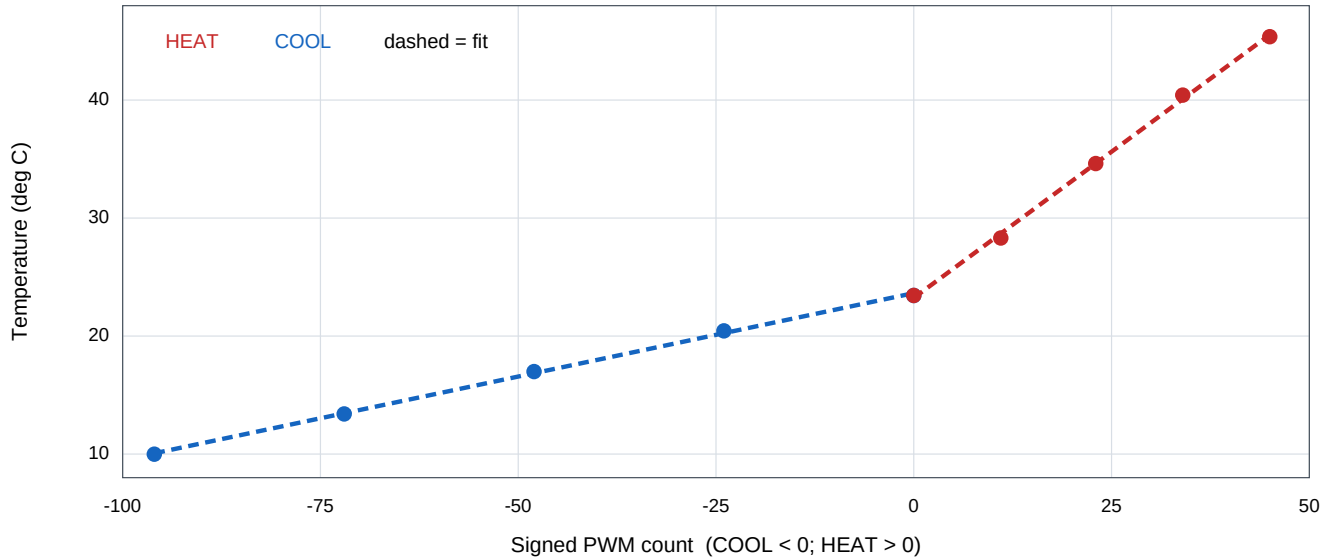


Figure 1. Five selected PWM magnitudes per direction; the two 0-PWM points overlap at 23.44 deg C. Steady temperatures are the means of the final approximately 20 s after the trace ceased sustained directional drift and fluctuated within a local band. Fits cover HEAT 0-45 and COOL -96-0 signed PWM counts.

HEAT slope	COOL slope	Measured ratio	Inferred Joule/Peltier
0.4954 deg C/count	0.1414 deg C/count	3.5035	0.5559

PWM averaging and steady-state energy balance

During one period, on-state current I flows for D times the period and zero current for the rest, where $D=|u|/255$. Direct integration gives $\langle I \rangle = DI$ and $\langle I^2 \rangle = DI^2$. In general $\langle I^2 \rangle$ is not $\langle I \rangle^2 = D^2 I^2$; the difference is $D(1-D)I^2$. Consequently, for fixed on-state current, both Peltier transport and Joule heating scale linearly with PWM duty, consistent with the nearly linear measured branches.

Let $d=u/255$ be signed duty and let Q_P and Q_J be positive full-on Peltier and object-face Joule heat rates. The reduced balance is $C \, dT/dt = Q_{TEC} - G(T-T_0)$, with $Q_{TEC} = dQ_P + |d|Q_J$. TEC conduction and other passive leaks are included once in G . At steady state the individual heat flows sum to zero: $G(T-T_0) = Q_{TEC}$.

Thus $T_h - T_0 = d(Q_P + Q_J)/G$ and $T_c - T_0 = d(Q_P - Q_J)/G$. Both signed-PWM slopes are positive: $m_h = (Q_P + Q_J)/(255G)$, $m_c = (Q_P - Q_J)/(255G)$. Their ratio gives $Q_J/Q_P = (r-1)/(r+1) = 0.5559$.

Manufacturer comparison and interpretation

Laird CP14-127-045 | hot-side temperature 27 deg C

Manufacturer specifications

Enter the values located in the 27 deg C hot-side table of the Laird CP14-127-045 data sheet. The condition and meaning of each value are shown below.

Specification	Meaning and operating condition	Value
R_M	Module electrical resistance at 27 deg C hot side	_____ ohm
$I_{\max}$	Maximum specified current at 27 deg C hot side	_____ A
$Q_{c,\max}$	Maximum cold-side pumping at Delta T = 0	_____ W
Delta T _{max}	Maximum no-load face temperature difference	_____ deg C

Source: Laird, *CP14-127-045 Thermoelectric Cooler*, 27 deg C hot-side specifications:
<https://sethfraden.github.io/Phys39F26-course/references/laird-tec-cp14-127-045.pdf>

Maximum-current calculation

At Delta T=0, passive TEC conduction vanishes. In the symmetric model, the object face receives half the total Joule heat. Therefore $Q_{J,\max} = 0.5 I_{\max}^2 R_M = \text{_____ W}$; $Q_{P,\max} = Q_{c,\max} + Q_{J,\max} = \text{_____ W}$; and $r_{\text{Laird},\max} = (Q_{P,\max} + Q_{J,\max}) / (Q_{P,\max} - Q_{J,\max}) = 1 + I_{\max}^2 R_M / Q_{c,\max} = \text{_____}$.

Comparison

The measured ratio is 3.5035; the manufacturer's maximum-current prediction is _____. Their difference is _____. Full PWM duty means continuously applying the H-bridge drive, not necessarily $I = I_{\max}$. Actual current depends on the supply settings, H-bridge and wiring drops, and module resistance. PWM rather than steady DC, finite face temperature difference, temperature-dependent properties, other passive paths, and fitting slightly curved data with one slope can also affect the comparison.

Passive conduction

When the block is hotter than room temperature, passive heat flows out; when colder, it flows in. The term $-G(T-T_0)$ therefore opposes both excursions. If G is approximately symmetric, conduction reduces both responses but does not by itself explain why the HEAT slope magnitude exceeds the COOL slope magnitude. Reversing current reverses Peltier transport, whereas Joule heating keeps the same sign.

Conclusion

Our open-loop measurements show an approximately linear steady-temperature response to PWM in each direction, with a larger heating susceptibility than cooling susceptibility. The fitted slope ratio is 3.50. Under the simplified near-room-temperature energy balance, that ratio corresponds to object-face Joule heating about 0.56 times the full-on Peltier heat rate. This is a model-based inference, not a direct measurement of either heat flow. Peltier transport reverses with current, while Joule heating keeps the same sign, so their effects add during heating and partly offset during cooling. Passive conduction carries heat away from a hot block and toward a cold one, opposing both departures. Manufacturer maximum-current values describe a different operating condition from our PWM-driven apparatus and should be compared on that basis.